Department of Microsystems Engineering

Probability and Statistics

PD Dr. Cristian Pasluosta

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Information and outline

Probability and Statistics

Lecture

- 11LE50V-6100
- Tuesday, 12^{00}-14^{00} am

Exercise

- 11LE50Ü-6100
- Friday, 12^{00}-14^{00} am (starting on Friday, November 20th)
- Thomas Krauskopf

Examination

- Written final exam → 100% of final mark
- You have to pass 50% of the exercises to be able to complete de module

Outline

Probability and Statistics	PD Dr. Cristian Pasluosta	WS 2026/2027	Thomas Krauskopf
Date	**Lecture**	**Date**	**Exercise**
Tuesday, 20.10.2026	Introduction	Friday, 23.10.2026	No exercise
Tuesday, 27.10.2026	Event Probabilities	Friday, 30.10.2026	No exercise
Tuesday, 03.11.2026	Random Numbers	Friday, 06.11.2026	No exercise
Tuesday, 10.11.2026	Expectation Values	Friday, 13.11.2026	No exercise
Tuesday, 17.11.2026	Large numbers, Central Limit Theorem	Friday, 20.11.2026	Introduction
Tuesday, 24.11.2026	Probabilitiy Densities	Friday, 27.11.2026	Excercise 1
Tuesday, 01.12.2026	Joint Distribution, Quantils, Moments	Friday, 04.12.2026	Excercise 2
Tuesday, 08.12.2026	MLE, MoM, Confidence Intervals	Friday, 11.12.2026	Excercise 3
Tuesday, 15.12.2026	Introduction Hypothesis Test	Friday, 18.12.2026	Excercise 4
Tuesday, 22.12.2026	T-Test	Christmas Holidays	
Christmas Holidays		Friday, 08.01.2027	Excercise 5
Tuesday, 12.01.2027	No lecture	Friday, 15.01.2027	Excercise 6
Tuesday, 19.01.2027	KGS, ANOVA	Friday, 22.01.2027	Excercise 7
Tuesday, 26.01.2027	Wilcoxon, Mann-Whitney, KruskalWallis	Friday, 29.01.2027	Excercise 8
Tuesday, 02.02.2027	Monte Carlo, RW, Samplesizes	Friday, 05.02.2027	Excercise 9
Tuesday, 09.02.2027	Exam Review	Friday, 12.02.2027	Excercise 10
Lecture: Tue. 12-14		**Exercise: Fr. 12-14**	

Preparation and wrapup

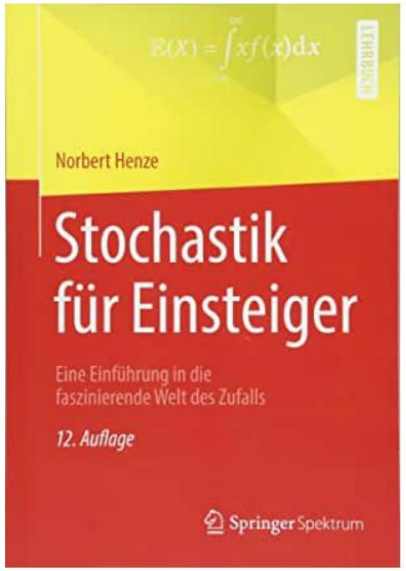
Literature (in general)

- I will upload lecture videos to **Ilias** portal (password: anova)
- literature in hardcopy or as PDF file
- some recommended books are available in the library

Consultation hours

- on appointment
- cristian.pasluosta@imtek.uni-freiburg.de
- thomas.krauskopf@imtek.uni-freiburg.de

Dennis' choice of textbooks:



Stochastik für Einsteiger: Eine Einführung in die faszinierende Welt des Zufalls
von Norbert Henze | 20. Juni 2018

Taschenbuch
27,99€

prime Lieferung bis **Mittwoch, 23. Oktober**
GRATIS Versand durch Amazon

Andere Angebote
16,19 € (30 gebrauchte und neue Artikel)

Kindle
19,99€



Mathematik für Ingenieure und Naturwissenschaftler Band 3: Vektoranalysis, Wahrscheinlichkeitsrechnung, Mathematische Statistik, Fehler- und Ausgleichsrechnung
von Lothar Papula | 15. März 2011

★★★★☆ 34

Taschenbuch
34,99€

2,94 € Versand
Nur noch 3 auf Lager

Andere Angebote
12,05 € (9 gebrauchte und neue Artikel)

Kindle
26,99€

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Dennis' choice of textbooks:



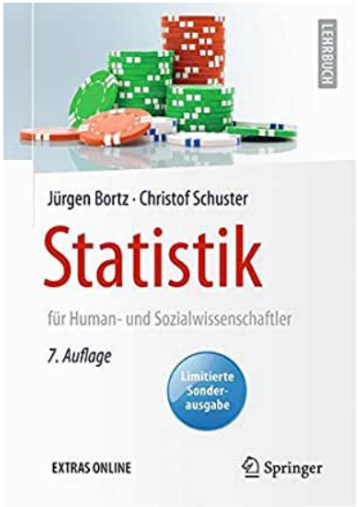
Einführung in Die Wahrscheinlichkeitstheorie und Statistik (Statistik und Ihre Anwendungen) (German Edition)
von Herold Dehling | 4. Mai 2004

★★☆☆☆ 4

Taschenbuch
39,99€

prime Lieferung bis **Mittwoch, 23. Oktober**
GRATIS Versand durch Amazon

Andere Angebote
9,46 € (6 gebrauchte und neue Artikel)



Statistik für Human- und Sozialwissenschaftler: Limitierte Sonderausgabe (Springer-Lehrbuch)
von Jürgen Bortz und Christof Schuster | 21. September 2016

★★★★☆ 19

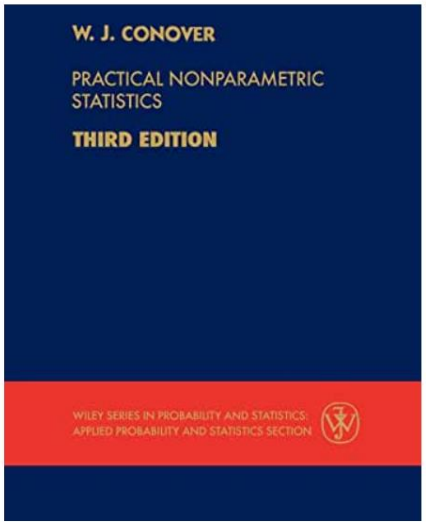
Taschenbuch
34,99€

prime Lieferung bis **morgen, 22. Oktober**
GRATIS Versand durch Amazon
Nur noch 19 auf Lager (mehr ist unterwegs).

Kindle
36,99€

Gebundenes Buch
54,99€

Dennis' choice of textbooks:



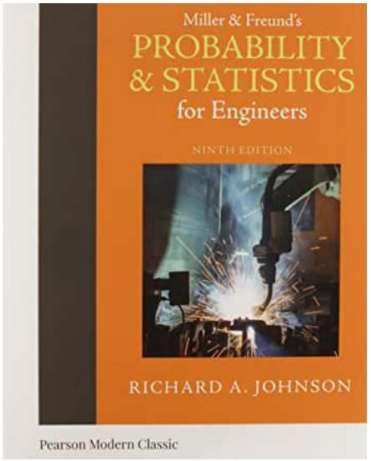
Practical Nonparametric Stats 3e (Wiley Series in Probability and Statistics)
von J. Conover, W. | 14. Dezember 1998
★★★★★ 1

Gebundenes Buch
159,00€

prime Lieferung bis **Mittwoch, 23. Oktober**
GRATIS Versand durch Amazon

Andere Angebote
155,99 € (9 gebrauchte und neue Artikel)

Taschenbuch
409,31€



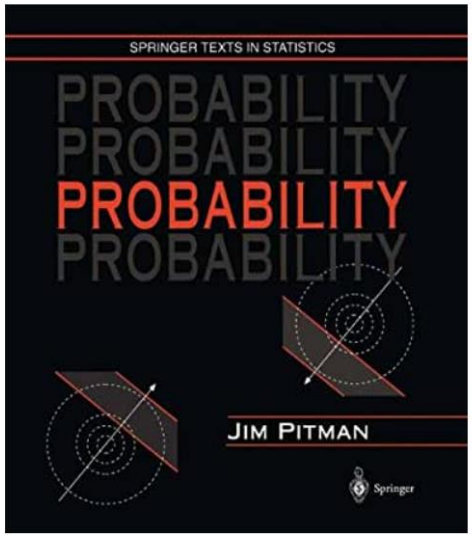
Miller & Freund's Probability and Statistics for Engineers (Pearson Modern Classics)
von Richard A. Johnson, Irwin Miller, et al. | 24. März 2018
★★★☆☆ 3

Taschenbuch
85,13€

Lieferung **Freitag, 25. Oktober – Montag, 28. Oktober**
7,04 € Versand
Nur noch 4 auf Lager (mehr ist unterwegs).

Gebundenes Buch
Andere Angebote
845,99 € (1 gebrauchter Artikel)

Literature (older)



Probability: Springer Texts in Statistics
von Jim Pitman | 1. Juni 1999
★★★★★ 3

Gebundenes Buch
51,23€

prime Lieferung bis **Mittwoch, 23. Oktober**
GRATIS Versand durch Amazon
Nur noch 1 auf Lager (mehr ist unterwegs).

Andere Angebote
39,63 € (12 gebrauchte und neue Artikel)

Kindle
35,86€



Statistik für Ingenieure: Wahrscheinlichkeitsrechnung und Datenauswertung endlich verständlich (Springer-Lehrbuch)
von Aeneas Rooch | 7. August 2014
★★★★★ 2

Taschenbuch
29,99€

KOSTENLOSE Lieferung
Nur noch 2 auf Lager

Andere Angebote
27,04 € (32 gebrauchte und neue Artikel)

Kindle
14,99€

Software Helpers

Software:

- R, Gretl (open source)
- Matlab, Statistics Toolbox
- Mathematica
- SPSS, StatSoft
- Origin
- Prism
- StatsMate (iPad)
- StatViz (iPad)
- CAS System: Axiom, Maple, Maxima, SageMath, SymPy
- MiniTab

Web Resources

http://www.statsoft.com/textbook/

http://stattrek.com/

http://www.r-project.org/

https://www.graphpad.com/data-analysis-resource-center/

https://www.real-statistics.com

https://sites.psu.edu/shinyapps/

IMTEK

Department of Microsystems Engineering

Intuition, Events and Probabilities

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Topics of the day:

Set Theory

- History and applications
- Definitions: Set, Sub-Set, Tupel, Element, Trial
- Calculation rules I: Algebraic rules
- Calculation rules II: Kolmogorov Axioms (1-3)

Probability

- Definitions: Probability P, probability function p, Partition, probability measure
- Laplace probability
- Odds and Ratio (Ratio-Checker)
- Distributions: Laplace, discrete uniform distribution

Combinatorics

- Ordered/unordered samples with/without replacement

Intuition

Contingency of things to happen... or not...

- A game of chance: lottery, roulette, etc.
- Daily life: rain and umbrella correlation...
- Economy: Stock market, Insurance
- Politics: Voting, Polls, Decision finding, to solve stalemate in the parliament in the state of RP, decision are made by drawing lots
- Science: A drop of ink in water (molecular view vs. comprehensive view)
- Science: Radioactive decay (individual C^{14} isotope, vs. a large mass of C^{14})
- Engineering: QM, MTBF,

Important recent applications

- **Medicine**: success of therapies at 95%
- **Biology**: planning and analysis of experiments
- **Economy**: marketing, stock market
- **Insurance** companies: premium calculation
- **Traffic**: queue, traffic jams
- **Opinion research**: projections during elections, campaigns, polls
- **Epidemiology**: disease spreading
- **Meteorology**: weather forecast.... 70% of rain or snow...
- **Gambling**: calculate the chances e.g. during poker or black jack, roulette
- **Engineering**: MTbF,
- **Physics**: Radioactive Decay, CERN, Manhatten Project, Apollo Program

History Basics



- Greek and Romans used dice from bones
- 1657 Christiaan Huygens wrote: ***Tractatus de Ratiociniis in Ludo Alea***, “If, in a game of chance the outcome is contingent, it is still possible to calculate the correlation between the likelihood of a player to win or to loose.”
- Pierre-Simon Laplace (1749-1827) definition of probability
- Andrey Nikolaevich Kolmogorov (1903-1987) set up probability theory
- Nobel prize on game theory 1994 Nash/Selten/Harsani

Mathematics of Chance

The Problem:

Certain “events” elude from becoming described by deterministic means.

Such an event is then called a “***stochastic phenomenon***”, since we do not have enough information to make a dedicated prediction.

Stochastic

Has become an independent field in mathematics (was part of physics before).

It includes both, the probability of events in a random experiment and the statistics of events in any experiment.

What Stochastic does?

Modeling of random events using mathematical terms.

Ideal random Experiment

A stochastic event is named "ideal random experiment" if the following conditions apply:

- The experiment is conducted under determined conditions (experimental conditions)
- The set of possible events is known prior to the implementation of experiments
- At least theoretically, the experiment can be repeated infinitively under the same conditions



Ronald A Fisher
(1890-1962)

Ronald A Fisher: The Design of Experiments, 1935

Definition: Set, Sub-Set and Element

- The set of all possible results in an experiment is called **Omega Ω**

- An individual possible result is called **element *w***

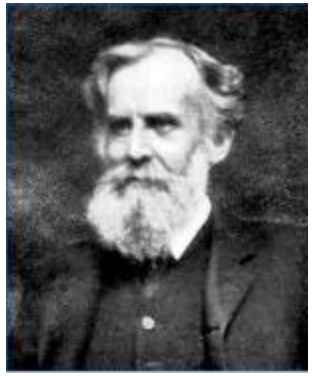
- Given two class of elements $A \in \Omega$ and $B \in \Omega$, if $A \subset B$,
  B is called a **superset of A** and A is a **subset of B**

- Let $A \in \Omega$ and $A_1, A_2, \ldots, A_n \in \Omega$. The subset $\{A_1, A_2, \ldots, A_n\}$ is called a **partition of A** if:
  $$A = \bigcup_{i=1}^{n} A_i \quad \textit{and} \quad A_i \cap A_j = \varnothing, \quad \textit{for } i \neq j \quad \text{holds.}$$

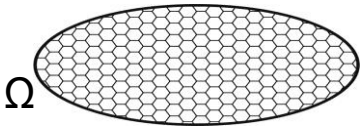
- If S = ({a,b,c}) then: P(S) = {{},{a},{b},{c},{a,b},{a,c},{b,c},{a,b,c}} is its **powerset**,
  i.e. the set of all subset including the empty set

- Let $A = \{1,2,3,4\} \Rightarrow |A| = 4$, 4 is the **cardinal number** of the set A

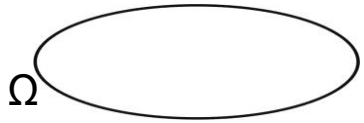
Theory of Sets in a Nutshell



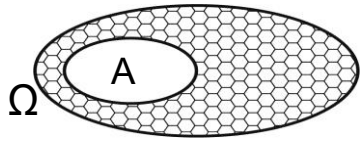
John Venn
(1834-1923)



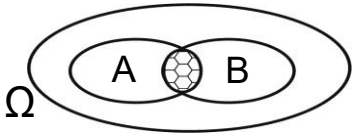
Ω sure event



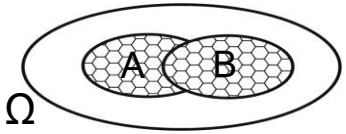
{} impossible event



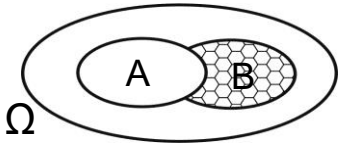
$\overline{A}$ complement of A (sometimes A^c)



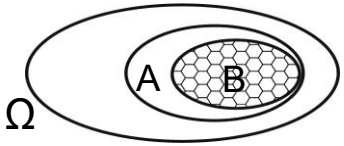
$A \cap B$ *A and B; iff w ∈ A and w ∈ B*



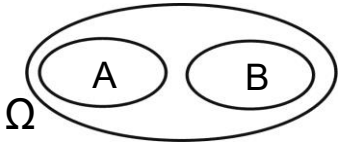
$A \cup B$ Union of A and B; *if w ∈ A or w ∈ B or both*



$B \setminus A$ B without A; *iff* $w \in B$ *and* $w \notin A$



$B \subset A$ B is a subset of A; $A \supset B$ A is superset of B



$A \cap B = \varnothing$ A and B are disjoint

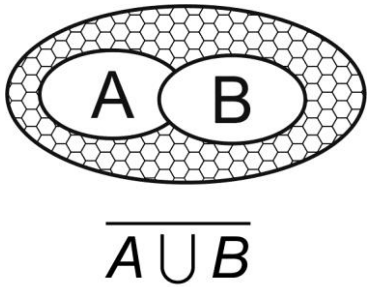
Set identities (algebraic calculation rules)

Commutative law: $A \cap B = B \cap A, \quad A \cup B = B \cup A$

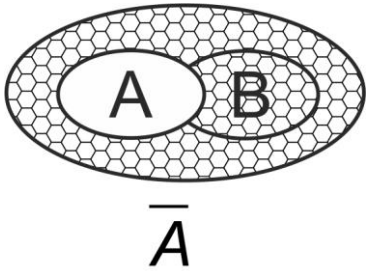
Associative law: $A \cap (B \cap C) = (A \cap B) \cap C, \quad A \cup (B \cup C) = (A \cup B) \cup C$

Distributive law: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

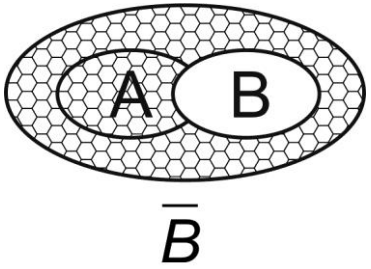
Law by de Morgan: $\overline{(A \cup B)} = \overline{A} \cap \overline{B}, \quad \overline{(A \cap B)} = \overline{A} \cup \overline{B}$



$\overline{A \cup B}$



$\overline{A}$



$\overline{B}$



Augustus De Morgan
(1806-1871)